

Highlights

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- Proposes a High-Dimensional Archive-based Differential Evolution algorithm (HDADE) for multimodal multi-objective optimization
- Introduces an archive-assisted evolutionary framework to balance convergence and diversity in high-dimensional decision spaces
- Designs a hybrid environmental selection strategy combining convergence assessment and diversity preservation
- Develops a cosine similarity-based diversity pruning mechanism for archive maintenance in high-dimensional spaces
- Validates superior performance on HDMMF and MMF benchmark problems against six state-of-the-art algorithms

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ABSTRACT

Multimodal multi-objective optimization problems (MMOP) are widespread in real-world applications, characterized by multiple Pareto optimal solutions (PSs) in the decision space that map to the same Pareto frontier (PF) in the objective space. Although current research on multimodal multi-objective optimization has matured considerably, existing algorithms often exhibit significant performance degradation when handling high-dimensional decision spaces. To address this challenge, this paper proposes a novel multimodal multi-objective evolutionary algorithm (MMEA) based on differential evolution framework. The algorithm employs an archive mechanism combined with an improved environmental selection strategy, introducing diversity archives to maintain solution distribution in the decision space while enhancing population exploration capabilities in high-dimensional decision spaces through improved environmental selection, thereby preserving more PSs. The integration of these two strategies effectively promotes balance between decision space diversity and objective space convergence, improving overall performance. Experimental results comparing the proposed algorithm with six advanced MMEAs across multiple benchmark problems validate its effectiveness.

1. Introduction

In practical applications, numerous complex systems require a balance among multiple interdependent performance indicators, a class of problems known as Multi-objective Optimization Problems (MOPs) [1, 2]. Unlike traditional single-objective optimization methods, MOPs are characterized by competitive relationships among objective functions, where it is typically impossible to find an ideal solution that simultaneously optimizes all objectives. Therefore, the solution to MOPs usually manifests as a set of solutions the Pareto optimal solutions. Within this solution set, any improvement in one objective function value will come at the expense of deterioration in at least one other objective function value. These Pareto optimal solutions, when mapped onto the objective space, form the Pareto Front (PF), which characterizes the trade-off relationships among various objectives and provides decision-makers with multiple feasible options.

It is noteworthy that multimodality, as an important characteristic of optimization problems, has not received sufficient attention in research. This feature manifests as several independent solutions distributed throughout the decision variable space that correspond to highly similar objective

function values. As shown in Fig. 1, despite solutions A, B, and C having significantly different positions in the decision space, their mappings in the objective space are remarkably close. Using path planning as an example, where A, B, and C represent three different routing strategies, certain important evaluation criteria (such as safety considerations) are difficult to quantify through explicit mathematical functions during actual decision-making processes. This leads to solutions with similar performance potentially exhibiting drastically different practical outcomes. For instance, solution A presents a longer path distance but offers the highest safety coefficient. In applications where safety is prioritized, even though solution A might be dominated by solutions B and C in the Pareto sense, decision-makers may still select it as the final choice. This phenomenon indicates that maintaining diversity within the solution set in the decision space has significant practical value when handling optimization problems that involve metrics that are difficult to quantify.

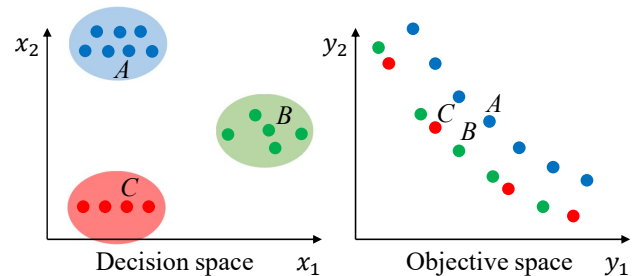


Figure 1: Illustration of multimodal multi-objective optimization problems (MMOPs), where there are two equivalent global PSs (region B and C) and one local PS (region A).

Furthermore, Multimodal Multi-objective Optimization Problems (MMOPs) in practical applications often involve high-dimensional decision spaces, where decision variables

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exhibit complex interdependence and coupling characteristics [3]. Such problems find widespread applications across multiple engineering and scientific domains. Taking feature selection in machine learning as an example, when facing high-dimensional datasets, the dimension of the decision space grows exponentially, making it difficult for conventional multimodal multi-objective algorithms to effectively identify and eliminate redundant features, thereby affecting the model's generalization performance [4, 5]. Similarly, in practical engineering applications such as transformer parameter optimization in power systems [6] and engine aerodynamic design in aerospace engineering [7], decision-makers need to seek a balance among multiple performance indicators within a high-dimensional parameter space.

However, research on MMOPs with high-dimensional decision spaces remains limited, and current MMOEAs struggle to solve high-dimensional optimization problems effectively. Distance metrics in high-dimensional spaces face two major challenges: the curse of dimensionality and sparsity, which cause traditional distance measurement methods to fail [6]. At present, common multimodal multi-objective optimization methods generally rely on distance metrics in the decision space. Specifically, crowding distance-based methods [8, 9] maintain population diversity by evaluating distance differences between individuals; clustering-based methods [10, 11] classify similar solutions into the same category based on spatial distances; niching-based methods [12, 13] identify and maintain different solution clusters using distance information; indicator-based methods [14, 15] must also consider distribution characteristics when evaluating solution quality; and optimization methods incorporating machine learning techniques [16, 17] similarly depend on accurate distance measurements to train models. However, when problem dimensionality increases significantly, these mechanisms generally face declining distance discrimination capability, making it difficult for algorithms to accurately estimate individual distribution density in the decision space. Therefore, it is necessary to explore diversity maintenance strategies that do not completely rely on traditional Euclidean distance for high-dimensional cases [18].

Current Multimodal Multiobjective Evolutionary Algorithms (MMOEAs) primarily focus on maintaining population diversity in the decision space while pursuing good convergence performance in the objective space, thereby preserving multiple equivalent Pareto optimal solutions [19]. However, when the dimensionality of the decision space is high, these algorithms often perform poorly in terms of convergence, struggling to consistently obtain high-quality solutions. On the other hand, existing large-scale multi-objective optimization techniques such as decision space partitioning methods based on variable correlation [20], problem transformation strategies that convert multi-objective problems into low-dimensional single-objective ones [21], dimensionality reduction techniques aimed at reducing search space dimensions, and novel evolutionary strategies designed to

enhance algorithm search capabilities [22, 23] can effectively mitigate challenges posed by high-dimensional decision variables but often lose critical information from the decision space during their operation, thereby weakening their capacity to maintain multiple equivalent Pareto optimal solutions.

To address this limitation, this paper proposes a High-Dimensional Archive-based Differential Evolution algorithm (HDADE) for multimodal multi-objective optimization. The algorithm maintains a diversity archive during the evolutionary process to assist the main population in achieving a balance between exploration and exploitation, thereby enhancing the quality of the final solution set. For environmental selection, this paper designs a hybrid strategy that comprehensively considers convergence and diversity: in the objective space, an individual's convergence performance is evaluated by calculating its distance to the ideal point; in the decision space, a locality-sensitive hashing method is introduced to assess the distributional diversity of individuals in randomly projected spaces. The main contributions of this research can be summarized in the following three aspects:

- An archive-assisted population evolution mechanism is proposed, significantly enhancing the algorithm's global search capability. By constructing a dynamically updated diversity archive and facilitating information exchange with the main population, collaborative evolution is achieved. This framework fully utilizes diversity information generated during the search process, thereby effectively avoiding premature convergence while improving convergence efficiency.
- An improved environmental selection strategy is designed, capable of effectively preserving multiple Pareto optimal solutions throughout the evolutionary process. This method evaluates individual convergence and diversity from both objective and decision spaces, thereby achieving a good balance between approximating the true Pareto front and maintaining behavioral diversity, ensuring that multiple global optimal solutions are simultaneously preserved.
- HDADE underwent systematic evaluation on standard test problems including HDMMF and MMF. Experimental results demonstrate that when handling high-dimensional multimodal optimization problems, HDADE significantly outperforms various existing state-of-the-art algorithms in terms of comprehensive performance, validating the effectiveness and robustness of the proposed strategies.

The remainder of this study is structured as follows. Section 2 briefly reviews related works in the literature. Subsequently, Section 3 details the proposed MSLS-MA methodology. Section 4 then presents an experimental evaluation of HDADE's performance, utilizing several multimodal evolutionary algorithms and HDMMF test suites. Following this, Section 5 provides a comprehensive analysis,

including an examination of parameter ε , the contributions of HDADe's subcomponents. Finally, Section 6 concludes the paper and outlines potential avenues for future research.

2. Preliminary work

2.1. Multi-objective optimization

Without loss of generality, the standard definition of multi-objective optimization is as follows:

$$\min f(x) = [f_1(x), f_2(x), \dots, f_n(x)] \quad (1)$$

$$\text{s.t. } g_i(x) \leq 0, \quad i = 1, 2, \dots, k; \quad (2)$$

$$h_j(x) = 0, \quad j = 1, 2, \dots, p. \quad (3)$$

In this context, x is the decision variable, $f(x)$ is the objective function, $f_i(x)$ represents the i -th objective function, and n denotes the number of objectives. A solution x_a is considered to Pareto dominate another solution x_b , if $\forall i = 1, 2, \dots, n \quad f_i(x_a) \leq f_i(x_b)$ and $\exists j = 1, 2, \dots, n \quad f_j(x_a) < f_j(x_b)$. Furthermore, a Pareto optimal solution is a solution that is not Pareto dominated by any other solution. In other words, within Pareto optimal solutions, no objective can improve its utility without compromising the utility of other objectives. The set of Pareto optimal solutions is called a Pareto set (PS). The image of the PS is known as the Pareto front (PF).

2.2. Multimodal optimization

Evolutionary algorithms for solving MMOPs can be viewed as processes of exploration and exploitation in the decision space [24, 25]. Generally, overemphasizing exploration can identify more feasible solutions but reduces convergence speed, while overemphasizing exploitation accelerates convergence but risks becoming trapped in local optima. To identify multiple optimal solutions in discrete optimization, researchers introduced niching techniques [26]. This approach divides the population into multiple subpopulations, each converging toward different potential optimal solutions, while maintaining diversity through specific mechanisms. Since Goldberg [27] first proposed the pre-selection strategy, various niching techniques have emerged. Crowding and fitness sharing strategies are particularly prevalent; crowding controls population updates by comparing similarities between offspring and parents, while fitness sharing reduces individual fitness based on proximity in the solution space, thereby regulating population density and maintaining diversity.

Additionally, multi-population strategies have been widely applied, such as the Species Conservation Genetic Algorithm (SCGA) [28]. Regarding clustering strategies, Qu et al. [29] proposed neighborhood mutation strategies (NCDE and NSDE) that execute DE operators within Euclidean neighborhoods. Cheng et al. [30] developed a Q-learning-based Brain Storm Optimization algorithm (QLBSO2N) featuring two niching strategies and learning-assisted evolution to enhance global search and convergence. Huang et al. [31] introduced an adaptive differential evolution algorithm

(WSNADE) that employs adaptive niching techniques and protective dual-scale mutation to improve diversity and search efficiency. Yan et al. [32] developed a knowledge-guided adaptive dynamic multimodal differential evolution algorithm that utilizes self-organizing maps for population clustering and dynamic response mechanisms. Ji et al. [33] proposed a heterogeneous ensemble surrogate-assisted interval multimodal particle swarm optimization algorithm (IMPSON-HES) with a surrogate model pool and incremental management to reduce function evaluations in costly optimization problems.

Despite the significant effectiveness demonstrated by existing methods in addressing multimodal multi-objective optimization problems, several limitations remain. First, as the number of global Pareto optimal solutions increases, existing methods struggle to consistently detect all optimal solutions. Second, when confronting high-dimensional and structurally complex optimization problems, these methods often fail to respond effectively due to a lack of targeted mechanisms.

To address mechanism failures in high-dimensional decision spaces, numerous innovative strategies have been proposed. For instance, Tian et al. [34] introduced guiding vectors to differentiate search directions among various subgroups by evaluating similarities between different subpopulations, specifically targeting large-scale sparse multimodal multi-objective problems. Xue et al. [35] adaptively adjusted solution convergence and diversity through guidance matrices, providing decision-makers with a series of high-quality and diversified matching schemes. Ding et al. [36] proposed a multi-population evolutionary algorithm based on hybrid hierarchical clustering, identifying equivalent Pareto solution sets through clustering and dividing populations, while designing an adaptive mutation strategy that integrates global and local guidance information to address optimization challenges with large-scale sparse Pareto fronts. Long et al. [37] further proposed a multimodal evolutionary algorithm based on dual similarity detection (SD-MMEA), combining online scoring of decision variables with a two-layer grouping strategy to enhance solving capabilities for high-dimensional sparse problems. Wu et al. [20] employed a fuzzy C-means clustering algorithm suitable for high-dimensional data to partition the search space into multiple subspaces, combining this with multi-stage optimization strategies to balance performance in both objective and decision spaces.

However, with the continuous increase in decision variable dimensionality, traditional multimodal multi-objective evolutionary algorithms may still face significant performance degradation due to the lack of specialized handling mechanisms. To address these challenges, this paper proposes a novel multimodal multi-objective evolutionary algorithm in the third section.

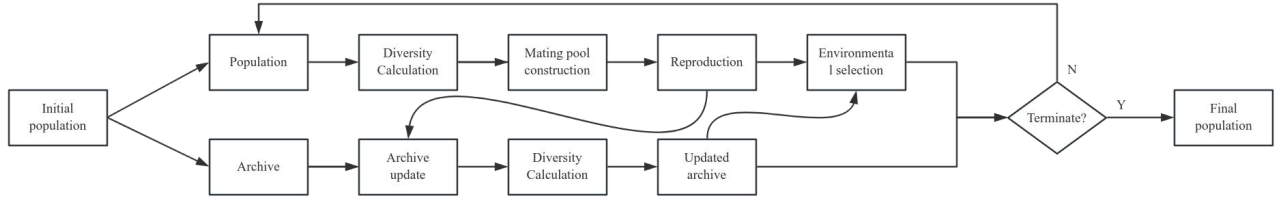


Figure 2: Diagram of the Archive-Assisted Evolutionary Framework for HDADE.

3. Proposed Method

3.1. Framework

Obtaining well-distributed multiple global optimal solutions is one of the core objectives in multimodal evolutionary algorithm research. To achieve this goal, existing methods typically incorporate distance metric mechanisms based on the decision space into their environmental selection strategies to maintain the distributional characteristics of solution populations across the set of optimal solutions. In low-dimensional optimization problems, such strategies can effectively locate a set of Pareto optimal solutions with comparable performance and uniform distribution. However, in high-dimensional scenarios, distance metrics face challenges such as the curse of dimensionality and data sparsity, causing significant performance degradation in traditional distance-based distribution assessment

Algorithm 1 General Framework of HDADE

Population size N

Input: Maximum function evaluations $MaxFEs$

Archive size $ArchiveSize$

Output: Final population P

```

1:  $Population \leftarrow Initialization()$ 
2:  $Archive \leftarrow Population$ 
3:  $FE = 0, ArchiveSize = N$ 
4: while  $FE \leq MaxFEs$  do
5:    $DS \leftarrow calculatediversityScore(Population)$ 
6:    $MatingPool \leftarrow TournamentSelection(DS)$ 
7:    $Off \leftarrow OperatorDE(Population, MatingPool)$ 
8:    $Pop \leftarrow [Population, Off]$ 
9:    $Archive \leftarrow UpdateArchive(Archive, Pop)$ 
10:   $Population \leftarrow$ 
       $EnvironmentalSelection(Pop, Archive)$ 
11:   $FE = FE + N$ 
12: end while
13: return  $Population$ 
  
```

Inspired by previous research [38, 39], this paper proposes a novel archiving-assisted evolutionary mechanism to better balance convergence and diversity in high-dimensional decision spaces. As illustrated in Figure 1, this mechanism first partitions the initialized population into archives and

then calculates their diversity in the high-dimensional decision space using the first norm. Based on this diversity measurement, the mechanism constructs mating pools, employs crossover and mutation operators for global search, and utilizes archives to preserve diverse individuals generated during the evolutionary process. By incorporating these diverse individuals from the archives into the environmental selection procedure, the algorithm enhances exploration of high-dimensional decision spaces. Furthermore, this approach shows potential for extension to other multi-modal evolutionary algorithms. Algorithm 1 presents the pseudocode of the entire HDADE framework.

3.2. Environmental selection

For multimodal multi-objective optimization problems, identical objective function values may correspond to individuals with significant differences in the decision space. Therefore, if a simple convergence-priority strategy is adopted to select solutions from the merged population, the algorithm will struggle to adequately explore the decision space and is likely to become trapped in local optimal solutions. Evidently, our goal is to retain solutions with superior performance while maintaining population diversity in high-dimensional decision spaces, but these objectives are often conflicting.

For high-dimensional decision space multi-objective optimization problems, this paper proposes an environmental selection mechanism for evolutionary algorithms designed to effectively maintain population diversity. The core of this mechanism consists of a series of diversity preservation strategies specifically designed for high-dimensional optimization problems, primarily including random subspace projection, reference vector-based environmental selection, and archive-assisted population update methods.

The random subspace projection selection mechanism addresses the "curse of dimensionality" in high-dimensional decision spaces by randomly selecting multiple low-dimensional subspaces (e.g., randomly selecting 3-5 dimensions to form a subspace) and calculating distances between individuals within each subspace to evaluate local diversity. By synthesizing diversity scores from multiple random subspaces, it effectively overcomes the "curse of dimensionality" in high-dimensional spaces, capturing multi-dimensional distribution characteristics that are difficult to compute directly

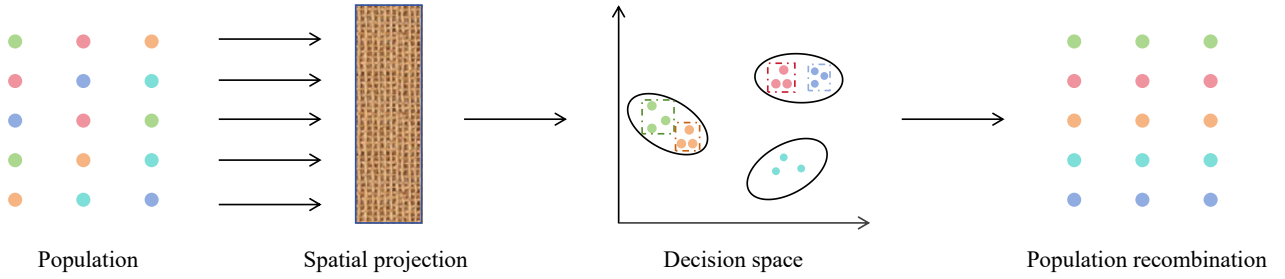


Figure 3: High-Dimensional Decision Space Projection and Similarity Mapping Framework.

Algorithm 2 Environmental Selection

Combined population P
Archive A
Input: Population size N
Archive retention size ϵ
Output: Selected population P

- 1: $Fronts \leftarrow \text{NDSort}(P.objs)$
- 2: $P \leftarrow \text{all individuals from } Front_1$
- 3: **while** $|P| < N$ **do**
- 4: $currentFront \leftarrow \text{next non-dominated front}$
- 5: **if** $|P| + |currentFront| \leq N$ **then**
- 6: $P \leftarrow P \cup currentFront$
- 7: **else**
- 8: $remaining \leftarrow N - |P|$
- 9: $selected \leftarrow \text{HYBRIDSELECTION}(currentFront, remaining)$
- 10: $P \leftarrow P \cup selected$
- 11: **end if**
- 12: **end while**
- 13: $P \leftarrow \text{DIVERSITYPRUNING}(P, N, \epsilon)$
- 14: $supplement \leftarrow \text{HYBRIDSELECTION}(A, N - |P|)$
- 15: $P \leftarrow P \cup supplement$
- 16: $P \leftarrow \text{remove duplicates from } P$
- 17: **Ensure** $|P| = N$
- 18: **return** P

in the complete high-dimensional space, thereby more accurately identifying individuals with unique positions in specific dimensional combinations.

The reference vector-based selection mechanism generates a set of uniformly distributed or coordinate axis-oriented reference vectors, maps each individual to the nearest reference direction, and evaluates its representativeness by calculating the projection length of the individual on the corresponding reference vector. In high-dimensional spaces, it combines random directions to ensure global exploration and coordinate axis directions to maintain dimensional coverage, thereby preserving directional diversity in the population while ensuring distributional breadth across key dimensions, providing a structured diversity assessment framework for the selection process.

In the archive-assisted population update mechanism, the hyperparameter ϵ first controls the retention scale of

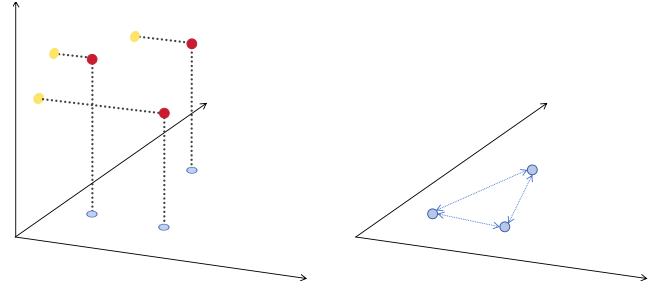


Figure 4: Random Subspace Diversity Projection of HDAE.

archived individuals. Its core lies in introducing a high-dimensional space pruning strategy based on cosine similarity, which effectively addresses high-dimensional challenges and ensures adequate expression of all key decision variables by replacing traditional Euclidean distance-based selection mechanisms.

The selection process adopts a hierarchical framework: first preserving non-dominated solutions, then applying specially designed diversity maintenance operators. Additionally, the algorithm incorporates multiple quality control mechanisms, including duplicate solution elimination, density-based individual pruning, and adaptive weighting strategies between convergence and diversity metrics. This environmental selection mechanism effectively mitigates the "curse of dimensionality" problem in evolutionary optimization by integrating locality-sensitive hashing concepts with reference vector-guided strategies. Consequently, this method is particularly suitable for complex practical optimization problems with high-dimensional decision spaces, where traditional diversity maintenance strategies often struggle to achieve ideal results. The environmental selection process is described in Algorithm 2.

3.3. Archive update

Archiving mechanisms assist population evolution by preserving and utilizing information generated during the evolutionary process, with the core aim of retaining diverse individuals produced during evolution, thereby enhancing algorithm performance. The key challenge of this mechanism lies in accurately identifying diverse individuals and

implementing effective retention. In existing research, various archive update strategies have been proposed to address this challenge. For instance, Peng et al. [40] proposed combining macro-archive sets with micro-archive sets, where macro-archive sets focus on maintaining population diversity during the global search process, while micro-archive sets concentrate on individual-level search processes, maintaining global diversity by eliminating poorly performing individuals during iterations, thereby simultaneously enhancing both convergence and diversity of the algorithm.

Existing archive update strategies are primarily suitable for low-dimensional decision spaces, as their diversity assessment typically relies on distance metrics. However, in high-dimensional decision spaces, local metric methods often fail, making it difficult to effectively maintain the diverse individuals generated during the evolutionary process. To address this issue, this paper proposes a diversity pruning mechanism based on cosine similarity, aimed at enhancing the diversity preservation capability of archive sets in high-dimensional optimization problems. The specific steps of the archive update process are shown in Algorithm 3.

Algorithm 3 Archive Update

Archive A
Input: Pop P
Archive size $ArchiveSize$
Output: Updated Archive A

- 1: **if** $Archive$ is empty **then**
- 2: $AllPop \leftarrow P$
- 3: **else**
- 4: $AllPop \leftarrow [A, P]$
- 5: **end if**
- 6: **if** $length(AllPop) > ArchiveSize$ **then**
- 7: $A \leftarrow DiversitySelection(AllPop, ArchiveSize)$
- 8: **else**
- 9: $A \leftarrow AllPop$
- 10: **end if**
- 11: **return** A

In the archive update process, cosine similarity is introduced as an evaluation criterion to measure the directional differences between individual solutions in the decision space, combined with convergence scores as an auxiliary screening basis. During the iterative pruning process, individuals with the most similar directions and poorer convergence are gradually eliminated, thereby effectively balancing solution diversity and overall quality in high-dimensional search spaces. Specifically, the decision variables of each individual are represented as vectors, and the cosine distance between vectors is calculated to quantify the degree of similarity between individuals.

$$\cos(\theta_{ij}) = \frac{\vec{x}_i \cdot \vec{x}_j}{\|\vec{x}_i\| \|\vec{x}_j\|} \quad (4)$$

Where $\vec{x}_i$ represents the decision vector of the i -th individual, $\vec{x}_j$ represents the decision vector of the j -th individual, $\|\vec{x}_i\|$ and $\|\vec{x}_j\|$ represent the Euclidean norms of

these two vectors, and $\vec{x}_i \cdot \vec{x}_j$ represents the dot product of the two vectors. Cosine similarity measures the difference between two individuals by calculating the cosine value of the angle between two vectors in a vector space. A cosine value approaching 1, with the angle approaching 0, indicates increasing similarity between the two vectors, while a cosine value approaching 0, with the angle approaching 90 degrees, indicates decreasing similarity between the two vectors. The detailed steps of diversity pruning are shown in Algorithm 4.

Algorithm 4 Diversity Selection

Population P
Input: Archive size $ArchiveSize$
Output: Selected archive A

- 1: $Decs \leftarrow dec(P)$
- 2: $nPop \leftarrow length(P)$
- 3: $keep \leftarrow [true, \dots, true]$
- 4: $minDissimilarity \leftarrow inf$
- 5: $dissimilarity \leftarrow cosineSimilarity(Decs)$
- 6: $convScores \leftarrow calConvergence(P)$
- 7: **while** $sum(keep) > ArchiveSize$ **do**
- 8: **if** $dissimilarity(i, j) < minDissimilarity$ **then**
- 9: $minDissimilarity \leftarrow dissimilarity(i, j)$
- 10: **end if**
- 11: Find pair (i, j) with minimal $dissimilarity$ among kept individuals
- 12: **if** $convScores[i] < convScores[j]$ **then**
- 13: $keep(i) \leftarrow false$
- 14: **else**
- 15: $keep(j) \leftarrow false$
- 16: **end if**
- 17: **end while**
- 18: **return** $P(keep)$

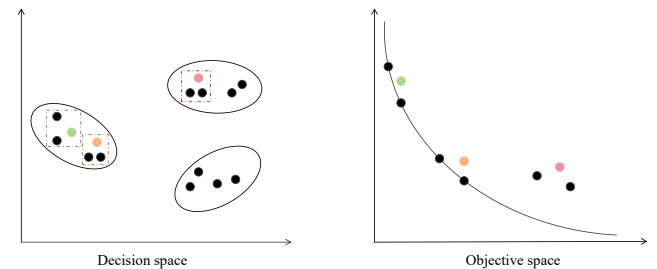


Figure 5: Similarity-Based Pruning in High-Dimensional Decision Space.

When numerous densely distributed individuals exist within the decision space, this paper proposes an archive update strategy based on cosine similarity and convergence assessment, aimed at prioritizing solutions that are widely distributed and exhibit superior convergence properties. This strategy first quantifies the structural similarity between individuals in the decision space through cosine similarity,

identifying groups of individuals with high similarity. Subsequently, convergence assessment is conducted for individuals within each group, analyzing their convergence performance in the objective space. Finally, individuals with overlapping convergence or inferior performance within groups are eliminated, retaining only representative solutions with optimal convergence characteristics. By integrating cosine similarity analysis with convergence performance screening, this method maintains population diversity in the decision space while ensuring archived individuals possess excellent convergence properties, thereby significantly reducing computational resource consumption while preserving the overall quality of the solution set.

4. Experiment

4.1. Experimental setting

4.1.1. Competitor algorithms and Benchmark

To evaluate the performance of the HDADE algorithm on high-dimensional multimodal multi-objective optimization problems, this study selected six effective evolutionary algorithms as benchmark comparison algorithms, including: HREA[41], HDMMOD[3], CoMMEA[42], TriMOEA-TA&R[43], MO_Ring_PSO_SCD[44], and MMEAWI[45].

For comprehensive and in-depth assessment of the HDADE algorithm's overall performance, this study selected two recognized multimodal multi-objective benchmark test sets for systematic experimental verification: the first being the HDMMF benchmark test set[3], and the second being the standard MMF test set[44]. The HDMMF test set comprises 15 problem instances with different characteristics, with problem construction designed to examine algorithms' ability to maintain population diversity in high-dimensional decision spaces while approximating the global Pareto front; the MMF test set encompasses 8 classical test instances that effectively examine algorithm performance in multimodal optimization under complex Pareto set structures and the uniformity of solution set distribution.

4.1.2. Performance metrics

For Multimodal Multi-objective Optimization Problems (MMOPs), commonly used evaluation metrics primarily include the Inverted Generational Distance in the Decision Space (IGDX) [46], which measures solution set diversity, and the Inverted Generational Distance (IGD) [47], which evaluates convergence. For a comprehensive understanding, these metrics are described as follows:

IGD metric: IGD measures the overall deviation between the obtained solution set and the true Pareto front (PF). A smaller IGD value indicates better convergence performance. The IGD calculation formula is as follows:

$$IGD(X) = \frac{1}{|X^*|} \sum_{y \in X^*} \min_{x \in X} \{ED(f(x), f(y))\} \quad (5)$$

where $ED(x, y)$ is the Euclidean distance between x and y . X and X^* represent the obtained solution set and a finite set

of Pareto optimal solutions uniformly sampled from the true PS.

IGDX metric: IGDX is a representative metric for evaluating the quality and diversity of solution sets in decision space for multimodal multi-objective algorithms. Its calculation formula is as follows:

$$IGDX(X) = \frac{1}{|X^*|} \sum_{y \in X^*} \min_{x \in X} \{ED(x, y)\} \quad (6)$$

where $ED(x, y)$ is the Euclidean distance between x and y .

4.2. Results

4.2.1. Performance on HDMMF test problems

To evaluate the performance of algorithms in solving high-dimensional multimodal multi-objective optimization problems, this paper conducted experimental research based on the standardized HDMMF test set. This test set comprises 15 test problems, and the experiments uniformly set the maximum number of function evaluations to 200,000 with a population size of 200 [3]. A summary of the experimental results is presented as follows.

The IGD metric quantifies an algorithm's convergence and diversity in the objective space, with lower values indicating better performance in terms of both convergence and diversity. This paper experimentally compared the proposed HDADE algorithm with six baseline algorithms across all 15 HDMMF test instances. As shown in Table 4, the HDADE algorithm demonstrated optimal stability and performance in most test instances, featuring high mean values and extremely low standard deviations, though it still faces risks of performance degradation on certain complex functions; HDMMODE and MMEAWI algorithms exhibited moderate performance, showing stability on simpler problems but considerable fluctuation on complex ones; the HREA algorithm demonstrated good robustness; while MO_Ring_PSO_SCD and CoMMEA algorithms, though possessing high flexibility and excelling in specific scenarios, showed weaker overall stability. Notably, all algorithms experienced significant performance degradation and sharp variance increases when handling extremely high-dimensional multimodal problems, reflecting that current algorithms' robustness still requires improvement when facing complex optimization spaces. This limitation may stem from the failure of distance measurement mechanisms in high-dimensional decision spaces, thereby affecting the algorithms' convergence processes toward optimal solutions.

The IGDX metric, as a key measure quantifying solution convergence and diversity in decision space, indicates better algorithm performance in decision space with lower values. According to the experimental results shown in Table 4, the six high-dimensional multimodal multi-objective optimization algorithms display distinct performance hierarchies on the IGDX metric: HDMMODE and HDADE algorithms demonstrated significant advantages, achieving the lowest mean values in most test problems, indicating their superior

Table 1

Comparison of IGD metric values for algorithms on HDMMF test cases

HDADE	HREA	HDMMODE	MMEAWI	TriMOEA-TA&R	MO_Ring_PSO_SCD	CoMMEA
4.4731e+0	1.2498e+3	6.3494e+0	1.2498e+3	1.2498e+3	8.5279e+0	1.2498e+3
1.8384e-2	8.1592e-2	4.5828e-2	6.9545e-2	1.3046e-1	5.5977e+2	8.8967e-1
6.8188e-2	1.6389e-2	7.0341e-3	4.0964e-3	8.2446e-3	8.0073e-2	9.3058e-3
5.6700e-3	7.4156e+1	1.1017e-2	1.7414e+1	2.0447e+1	3.5709e+2	3.0486e+1
1.2166e+9	6.0738e+14	1.0511e+8	9.2762e+11	4.3850e+32	2.3715e+7	4.2883e+8
3.5959e+1	3.1305e+8	2.4431e+0	6.5743e+1	4.7459e+0	2.3477e+1	5.4718e-2
6.1418e-2	8.7800e+14	2.0008e-2	1.2311e+1	4.0332e+0	1.4697e+2	8.7420e+0
1.0303e+0	8.4942e-1	6.8250e-1	1.4713e-1	6.1698e+0	2.0071e+0	7.0150e-1
5.1849e-1	4.7454e+4	2.5207e-1	4.7454e+4	4.7454e+4	7.4402e+0	4.7454e+4
7.4214e-3	8.3750e+2	2.3453e-2	8.3733e+2	8.3733e+2	6.3464e+3	8.3733e+2
4.0325e+2	4.9927e+3	1.8851e+3	4.8186e+3	4.7864e+3	5.2495e+2	4.7990e+3
5.4693e+1	1.5045e+3	5.7398e+1	1.4369e+3	1.4329e+3	2.7173e+2	1.4314e+3
6.1372e+3	4.0101e+4	2.5714e+4	4.0091e+4	4.0087e+4	3.1358e+6	4.0087e+4
8.0331e-3	1.3927e+4	5.5908e-3	1.4245e+4	1.3927e+4	4.2428e+2	1.3927e+4
8.0466e+0	8.0487e+2	8.0702e+0	6.2651e+3	8.0487e+2	1.1690e+1	8.0487e+2

ability to approximate the true Pareto front while maintaining diversity in decision space; HREA and MMEAWI algorithms, though stable in some test instances, showed higher overall IGD means, indicating relatively inferior performance; TriMOEA-TA&R, MO_Ring_PSO_SCD, and CoMMEA algorithms performed at moderate levels, showing neither consistent advantages nor obvious disadvantages across the test set. Statistical analysis further confirmed the significance of HDMMODE and HDADE's advantages in IGD performance, especially in high-dimensional problem spaces.

The fundamental reason for differences in IGD performance among high-dimensional multimodal multi-objective optimization algorithms lies in their different mechanisms for addressing the "high-dimensional space distance metric failure" problem. HDMMODE's excellent performance is primarily attributed to its innovative approach that avoids the limitations of direct reliance on Euclidean distance measurements, instead implicitly handling relationships between decision variables through a dual-population co-evolutionary framework and differential evolution operators, effectively overcoming the "distance concentration phenomenon" in high-dimensional spaces; the HDADE algorithm, through its carefully designed population-archive co-evolutionary framework and diversity-oriented archive update mechanism, achieves efficient processing of high-dimensional decision variables, reducing dependence on global distance metrics, thereby more accurately identifying and maintaining equivalent solution sets in high-dimensional spaces. In comparison, HREA, MMEAWI, CoMMEA, MO_Ring_PSO_SCD, and TriMOEA-TA&R algorithms, though each with innovative features, generally rely on Euclidean distance-based diversity maintenance mechanisms, which inherently face reliability challenges in high-dimensional spaces as dimensions increase,

the differences between pairwise distances gradually diminish and tend toward equality (the "distance concentration phenomenon"), causing distance-based discrimination mechanisms to fail, preventing algorithms from accurately identifying and maintaining diversity structures in decision space, and ultimately performing poorly in IGD metric evaluations. This finding provides important insights for high-dimensional multi-objective optimization algorithm design: effective high-dimensional algorithms should reduce dependence on explicit distance calculations and instead adopt implicit diversity maintenance strategies better suited to high-dimensional characteristics.

Experimental results demonstrate that the HDADE algorithm exhibits significant advantages in high-dimensional multi-modal multi-objective optimization problems. In performance evaluations, HDADE achieved the best results in 7 out of 15 standard test cases for the IGD metric, outperforming comparative algorithms; for the IGD metric, HDADE obtained the best performance in 6 test cases, tied with the HDMMODE algorithm for first place, confirming the algorithm's dual strengths in maintaining convergence in objective space and diversity in decision space. Notably, even in test cases where HDADE did not achieve optimal performance, its metric values typically remained close to the optimum, demonstrating high algorithm stability and robustness; in contrast, the HREA, MMEAWI, TriMOEA-TA&R, and CoMMEA algorithms performed relatively poorly in most test cases. Comprehensive experimental results indicate that the HDADE algorithm effectively approximates the true Pareto front in both objective and decision spaces, exhibiting exceptional convergence and diversity in solving high-dimensional multi-modal multi-objective optimization problems, thus providing an efficient solution method for this field.

Table 2Comparison of IGD_X metric values for algorithms on HDMMF test cases

HDADE	HREA	HDMMODE	MMEAWI	TriMOEA-TA&R	MO_Ring_PSO_SCD	CoMMEA
1.0202e+1	1.2247e+1	1.0253e+1	1.2247e+1	1.2247e+1	1.2974e+1	1.2247e+1
5.2202e-2	4.8280e+0	8.7129e-2	5.1404e+0	4.6191e+0	1.8984e+0	3.3858e+0
1.6747e+0	4.2737e+0	2.2558e+0	7.4186e+0	6.4775e+0	4.6695e+0	7.4170e+0
3.7721e+0	2.3497e+1	3.7742e+0	2.3088e+1	1.1356e+1	6.8531e+0	1.9141e+1
2.2027e+1	4.1689e+1	2.2119e+1	3.6799e+1	3.3461e+1	3.4413e+1	3.0674e+1
1.4277e+1	1.9029e+1	1.3643e+1	3.2293e+1	1.7397e+1	1.5085e+1	2.0555e+1
1.1982e+1	4.0963e+1	1.0207e+1	2.7952e+1	1.6499e+1	2.4190e+1	1.6839e+1
3.9239e+0	1.0298e+1	4.9199e+0	1.1142e+1	9.6088e+0	5.9226e+0	5.3766e+0
2.3519e+0	2.9901e+0	1.9095e+0	2.9901e+0	2.9901e+0	3.0560e+0	2.9901e+0
3.8753e+0	8.0503e+0	2.1007e+0	8.0548e+0	8.0636e+0	6.4528e+0	8.0636e+0
1.2553e+1	1.4884e+1	1.3986e+1	1.4888e+1	1.4888e+1	1.3702e+1	1.4887e+1
1.2685e+1	1.2216e+1	1.0900e+1	1.2220e+1	1.2220e+1	1.1261e+1	1.2219e+1
1.9995e+1	2.7950e+1	1.9052e+1	2.7950e+1	2.7950e+1	3.2017e+1	2.7950e+1
4.6902e+0	8.3730e+0	5.0691e+0	8.3730e+0	8.3814e+0	1.2029e+1	8.3730e+0
1.3227e+1	1.0462e+1	1.5036e+1	1.0495e+1	1.0484e+1	2.0520e+1	1.0469e+1

Table 3

Comparison of IGD metric values for algorithms on MMF test cases

HDADE	HREA	HDMMODE	MMEAWI	TriMOEA-TA&R	MO_Ring_PSO_SCD	CoMMEA
4.3474e-3	3.7323e-3	5.8734e-3	3.5382e-3	4.4913e-3	4.5401e-3	4.1296e-3
1.1194e-2	2.7784e-2	1.4881e-2	1.1231e-2	2.6661e-2	3.4919e-2	1.7252e-2
8.9167e-3	1.3119e-2	1.3439e-2	1.0523e-2	2.6116e-2	4.0238e-2	1.1685e-2
4.9276e-3	3.2776e-3	6.2836e-3	3.8678e-3	6.6955e-2	4.1952e-3	4.3994e-3
4.2201e-3	3.6098e-3	6.2235e-3	3.3936e-3	6.2227e-3	3.6684e-3	3.9653e-3
4.7714e-3	3.5409e-3	5.5532e-3	3.5025e-3	3.8688e-3	3.8395e-3	4.7056e-3
5.4631e-3	3.6128e-3	5.9172e-3	3.8877e-3	3.5763e-3	3.7963e-3	3.5164e-3
4.4669e-3	3.9764e-3	7.0008e-3	3.8584e-3	3.3893e-3	4.5360e-3	6.1734e-3

4.2.2. Performance on MMF test problems

To evaluate algorithm performance in solving multi-modal multi-objective problems, this research conducted experiments based on the standardized MMF benchmark test set. This test set comprises 5 test instances, with experimental settings uniformly setting the maximum function evaluations to $5000 \times D$ and population size to $100 \times D$, where D represents the dimension of decision variables [48]. The experimental results are summarized as follows:

In the comprehensive evaluation of MMF test cases, the IGD and IGD_X metrics revealed distinct performance stratification phenomena. Regarding the IGD metric, MMEAWI demonstrated the most outstanding performance, consistently presenting optimal performance across multiple test groups; HREA and HDADE followed, performing well on specific problems but exhibiting fluctuations; HDMMODE, CoMMEA, TriMOEA-TA&R, and MO_Ring_PSO_SCD showed relatively weaker IGD performance. However, IGD_X metric evaluations presented a different performance distribution: HDADE, HREA, and CoMMEA held advantages in decision space diversity, while the remaining four algorithms performed inferiorly on this metric, indicating a

significant trade-off between objective space convergence and decision space diversity in optimization algorithms.

The significant differences in algorithm performance stem from the balance of strengths and weaknesses in their core mechanism designs. MMEAWI successfully achieved balance between objective space convergence and decision space diversity through carefully designed weighted indicators and dual archive strategies, thus demonstrating outstanding performance on the IGD metric; HREA's hierarchical ranking mechanism effectively enhanced population diversity, but its complex computational process led to efficiency fluctuations; HDADE strengthened convergence performance through archive-assisted evolutionary strategies, but consumed evaluation counts too rapidly; HDMMODE's dual-population architecture enhanced decision space exploration while sacrificing some objective space convergence efficiency; TriMOEA-TA&R's recombination strategy reduced objective space convergence precision due to introduced noise; while MO_Ring_PSO_SCD's ring topology, though maintaining local information exchange, limited global information propagation. These mechanism characteristics directly determined performance differences of each

Table 4

Comparison of IGDX metric values for algorithms on MMF test cases

HDADE	HREA	HDMMODE	MMEAWI	TriMOEA-TA&R	MO_Ring_PSO_SCD	CoMMEA
5.3789e-2	3.8894e-2	6.6702e-2	4.3856e-2	7.6620e-2	6.4612e-2	3.8592e-2
1.6741e-2	4.8165e-2	3.2637e-2	3.3766e-2	6.4636e-2	1.1713e-1	3.3816e-2
1.5424e-2	3.0112e-2	3.2133e-2	2.5785e-2	7.0934e-2	6.2425e-2	2.2072e-2
4.5426e-2	1.9312e-2	3.8198e-2	2.5135e-2	1.6173e-1	4.4271e-2	2.4095e-2
8.5254e-2	6.3508e-2	1.0571e-1	7.6976e-2	1.0811e-1	1.1088e-1	5.8246e-2
8.3574e-2	5.5501e-2	8.3343e-2	6.3040e-2	8.9214e-2	9.4698e-2	5.5014e-2
3.8948e-2	1.9503e-2	3.8329e-2	2.7508e-2	3.4742e-2	3.9788e-2	2.2832e-2
6.3389e-2	4.6664e-2	1.0146e-1	9.5748e-2	2.6489e-1	9.7279e-2	5.2595e-2

algorithm under different evaluation metrics, reflecting inherent challenges in multi-objective optimization algorithm design.

In the field of multimodal multi-objective optimization, differences in algorithm performance primarily arise from trade-offs among key capabilities in their core mechanism designs. Specifically, the MMEAWI algorithm effectively balanced convergence in objective space and diversity in decision space through carefully constructed weighted indicators and dual archive strategies, thus performing excellently on the Inverted Generational Distance (IGD) metric. While the hierarchical sorting mechanism adopted by the HREA algorithm helped enhance population diversity, its higher computational complexity led to certain fluctuations in operational efficiency. The HDADE algorithm improved convergence capability through archive-assisted differential evolution strategies but affected overall performance by depleting evaluation counts too early. HDMMODE's dual-population structure strengthened decision space exploration capability while weakening convergence efficiency in objective space. TriMOEA-TA&R introduced a certain degree of noise through individual recombination strategies, thus reducing convergence precision in objective space. Meanwhile, the ring topology structure adopted by MO_Ring_PSO_SCD, though beneficial for maintaining local information interaction, limited effective global information propagation. These mechanism characteristics directly influenced the performance of various algorithms under different performance metrics, reflecting the prevalent trade-off challenges in multimodal multi-objective optimization algorithm design.

Based on comprehensive analysis of IGD and IGDX metrics in low-dimensional MMF test cases, results demonstrate that HDADE exhibits excellent problem-solving capabilities in low-dimensional contexts, consistently outperforming HDMMODE across all test instances with lower metric values in both convergence and diversity measurements. Although HDADE does not achieve optimal results in some individual low-dimensional test cases compared to certain algorithms in the comparison set, its overall performance remains competitive, validating the algorithm's effectiveness in low-dimensional spaces. Particularly noteworthy is that, as an algorithm primarily designed for

high-dimensional multi-modal multi-objective optimization problems, HDADE performs remarkably well in low-dimensional problems, demonstrating its versatility and adaptability. This robust performance across problems of varying dimensionality, especially the enhanced convergence capability and decision space diversity maintenance already evident in low-dimensional problems, provides a solid foundation for the algorithm's application in more challenging high-dimensional multi-modal multi-objective optimization scenarios, while also indicating that HDADE is a universal algorithm with outstanding performance across optimization problems of different dimensionalities.

5. Further Discussion

5.1. Analysis on parameter ϵ

We conducted experimental research on parameter settings for the HDADE algorithm. This section elaborates in detail on the configuration scheme for the archive retention parameter and its impact on algorithm performance. Experimental results indicate that when setting the archive retention parameter ϵ to 0.8, the algorithm can retain individuals with good convergence properties in the main population while maintaining sufficient diversity through the archive population, thereby avoiding degradation of the population's overall convergence performance due to excessive diversification.

Table 5 illustrates the impact of the archive retention parameter ϵ on algorithm performance. Analysis of the tabulated data indicates that the optimal value range for ϵ is [0.7, 0.9]. For simpler test problems, the setting of ϵ does not significantly influence solution results. This is primarily because the problem dimensionality is lower, and even when the algorithm places excessive emphasis on diversity individuals retained in the archive, its global search operations can still effectively locate all global optimal solutions. In contrast, for more complex test problems, the value of ϵ significantly affects solution quality: excessively low ϵ values impair exploitation capability, hindering the discovery of potential optimal regions; while excessively high ϵ values diminish exploration capability, preventing the algorithm from fully converging to global optimal solutions. Notably, on the HDMMF test problem set, the algorithm still obtains

Table 5
Archiving Threshold ϵ -Parameter Sensitivity Analysis

HDADE-($\epsilon=0.5$)	HDADE-($\epsilon=0.6$)	HDADE-($\epsilon=0.7$)	HDADE-($\epsilon=0.8$)	HDADE-($\epsilon=0.9$)
1.2271e+1	1.2668e+1	1.6723e+1	1.0202e+1	1.0613e+1
5.2313e-2	1.5313e+0	5.3456e-2	5.2202e-2	5.4639e-2
1.5536e+0	1.6527e+0	1.6522e+0	1.6747e+0	1.4456e+0
3.7734e+0	3.7785e+0	3.7780e+0	3.7721e+0	3.7760e+0
2.3474e+1	2.6552e+1	2.1907e+1	2.2027e+1	2.4551e+1
1.4073e+1	1.4183e+1	1.4379e+1	1.4277e+1	1.3723e+1
1.1891e+1	1.3494e+1	1.2850e+1	1.1982e+1	1.1555e+1
4.1865e+0	3.5727e+0	4.2136e+0	3.9239e+0	3.9416e+0
2.0940e+0	2.5560e+0	2.3360e+0	2.3519e+0	2.7391e+0
3.5142e+0	4.7751e+0	2.6581e+0	3.8753e+0	3.8904e+0
1.2315e+1	1.2924e+1	1.2842e+1	1.2553e+1	1.1675e+1
1.3103e+1	1.2521e+1	1.1855e+1	1.2685e+1	1.3293e+1
2.0284e+1	1.9977e+1	1.9017e+1	1.9995e+1	2.0009e+1
4.8866e+0	4.6715e+0	4.7264e+0	4.6902e+0	4.7867e+0
1.3855e+1	1.3475e+1	1.4556e+1	1.3227e+1	1.4496e+1

relatively good solutions even with suboptimal parameter settings.

5.2. Analysis on subcomponents of HDADE

To systematically investigate the impact of key algorithmic components on performance efficiency, we developed two variants of the HDADE algorithm: HDADE-V1 and HDADE-V2. These variants were designed through controlled component elimination. In HDADE-V1, we replaced the random subspace projection selection mechanism with a grid-based decision space selection approach. For HDADE-V2, we further eliminated the archive-assisted population update mechanism, thereby ensuring that all individuals retained during the environmental selection process are exclusively derived from the joint population. This methodical decomposition allows for quantitative assessment of each component's contribution to the algorithm's overall effectiveness.

Figure 6 illustrates the impact on algorithm performance after removing specific components. Analysis of the IGD_X metric data reveals significant insights into component contributions within the HDADE algorithm family. The original HDADE algorithm demonstrates superior performance in 9 out of 15 test cases, while HDADE-V1 and HDADE-V2 achieve optimal results in only 4 and 2 cases, respectively. This performance distribution suggests that both key components substantially enhance algorithmic efficiency. The replacement of random subspace projection with grid-based selection in HDADE-V1 generally diminishes performance, although it yields improvements in specific problem structures (4 test cases). Moreover, the further elimination of the archive-assisted population update mechanism in HDADE-V2 leads to additional performance degradation, with this variant achieving superior results in only 2 instances. This systematic degradation pattern confirms that both components contribute meaningfully to HDADE's effectiveness,

Table 6
Component Configuration of HDADE and Its Variant Algorithms

HDADE-v1	HDADE-v2	HDADE
1.4344e+1	1.0619e+1	1.0202e+1
5.3366e-2	4.5437e-2	5.2202e-2
1.8155e+0	1.8314e+0	1.6747e+0
3.8117e+0	3.7789e+0	3.7721e+0
2.4169e+1	2.4602e+1	2.2027e+1
1.2826e+1	1.3831e+1	1.4277e+1
1.0617e+1	1.1499e+1	1.1982e+1
4.1505e+0	4.4464e+0	3.9239e+0
2.2335e+0	2.0350e+0	2.3519e+0
2.6777e+0	6.9948e+0	3.8753e+0
1.4016e+1	1.3607e+1	1.2553e+1
1.1479e+1	1.2385e+1	1.2685e+1
2.0590e+1	2.0522e+1	1.9995e+1
5.6811e+0	5.8684e+0	4.6902e+0
1.3850e+1	1.3985e+1	1.3227e+1

with the archive-assisted mechanism potentially exerting a somewhat stronger influence, as evidenced by the more pronounced performance decline observed in HDADE-V2 compared to HDADE-V1.

6. Conclusion

This paper proposes an archive-assisted differential evolution algorithm for solving high-dimensional multi-modal multi-objective optimization problems, which effectively integrates exploration and exploitation processes to simultaneously locate multiple optimal solutions. By leveraging a diversity archive mechanism, the algorithm harmonizes population convergence in the objective space and exploration capabilities in the decision space. To further

enhance algorithmic performance, two innovative mechanisms are designed: 1) Random Subspace Projection Selection Mechanism: By randomly sampling multiple low-dimensional subspaces within the high-dimensional space and comprehensively evaluating individual distributions, the mechanism achieves precise capturing of local diversity; 2) Archive-Assisted Population Update Mechanism: Utilizing hyperparameter control for archive scale and implementing a cosine similarity-based pruning strategy to replace traditional archive selection approaches, the mechanism ensures effective preservation of historical elite individuals and comprehensive expression of critical decision variables in high-dimensional environments.

To verify the efficiency and effectiveness of this algorithm, HDADE was comparatively evaluated against six popular multimodal multi-objective evolutionary algorithms. HDADE demonstrated significant advantages on two key metrics (IGD and IGDX). Additional experiments on MMF test instances further confirmed the superiority of HDADE in obtaining better and more stable solutions.

Future research directions include: 1) Scalability enhancement: HDADE's performance on large-scale instances requires improvement. Establishing large-scale high-dimensional multimodal multi-objective benchmarks is crucial for effectively evaluating algorithms in complex scenarios; 2) Metric improvement: Current metrics rely on sampling points and fail to fully capture solution set characteristics (e.g., differences between solutions, diversity of feasible solutions). A comprehensive evaluation framework is urgently needed; 3) Multimodal mechanism optimization: Further research on adaptive diversity maintenance strategies and population architectures in high-dimensional decision spaces can improve the efficiency of multimodal optimization.

CRedit authorship contribution statement

Lida Zhang: Writing review & editing, Writing original draft, Methodology. **Yan Zheng:** Writing original draft. **Xingyi Yao:** Writing original draft. **Wenhua Li:** Writing review & editing, Supervision, Methodology, Conceptualization. **Rui Wang:** Resources & Methodology. **Tao Zhang:** Writing review & editing.

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